

1 Text Classification: Sparse Features vs Contextual Embeddings

1. Introduction and Dataset

This project explores and compares classical machine learning models on two fundamentally different text representations: **sparse statistical features** (Bag-of-Words, TF-IDF) and **dense contextual embeddings** (DistilBERT). Pretrained deep learning models are used strictly as feature extractors — no fine-tuning is performed.

Dataset: AG News

A multi-class news classification benchmark from Hugging Face.

Classes: 4 topics — World, Sports, Business, Sci/Tech.

Train: 8,000 samples **Test:** 2,000 samples.

Articles are short paragraphs; classes are approximately balanced. Sampled to maintain computational feasibility while ensuring statistical significance.

Project Objective

Evaluate five classical ML models across two representation paradigms:

- Bag-of-Words (BoW) — raw term frequency
- Bigram TF-IDF — weighted, context-aware sparse features
- DistilBERT [CLS] — dense 768-dim contextual embeddings

Assess the effect of dimensionality reduction (SVD, PCA) on each.

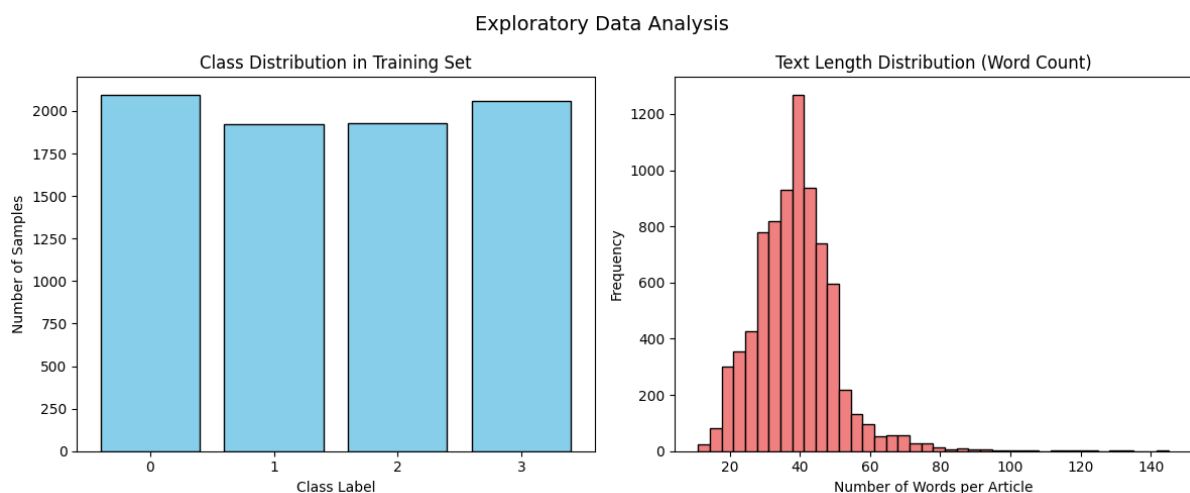


Figure 1: EDA

2. Methodology

Part A — Sparse Feature Representations

Raw text was processed through a standard preprocessing pipeline before feature extraction:

Unigram Bag-of-Words (BoW)

Captures raw frequency of individual words.

Preprocessing:

- Lowercase tokenisation
- Stop-word removal via `ENGLISH_STOP_WORDS`
- Vocabulary pruned to top `max_features=5000`

Produces a sparse $n \times 5000$ count matrix. High-dimensional but fast to compute.

Bigram TF-IDF

Captures term frequency-inverse document frequency for unigrams and bigrams (`ngram_range=(1,2)`), penalising highly frequent, uninformative words.

Key formula:

$$\text{TF-IDF}(t, d) = \text{tf}(t, d) \cdot \log \frac{N}{\text{df}(t)}$$

Captures basic word-pair context; outperforms BoW on most tasks due to term weighting.

Part B — Contextual Embeddings (DistilBERT)

DistilBERT Feature Extraction Pipeline

Model: `distilbert-base-uncased` — frozen, no fine-tuning.

Extraction: Tokenised text is passed through the model. The [CLS] token embedding from the last hidden state (`outputs.last_hidden_state[:, 0, :]`) serves as a 768-dimensional sentence-level representation.

$$\mathbf{e}_i = \text{DistilBERT}(x_i)_{[\text{CLS}]} \in \mathbb{R}^{768}$$

Caching: Embeddings computed once and saved as `.npy` arrays to avoid redundant GPU computation across runs.

Part D — Dimensionality Reduction

Truncated SVD on TF-IDF

Reduces the sparse $n \times 5000$ TF-IDF matrix to 100 components — analogous to Latent Semantic Analysis (LSA).

$$X \approx U_k \Sigma_k V_k^T, \quad k = 100$$

Effect: Slight drop in absolute accuracy vs. full 5000-dim space, but significantly faster training for SVM and KNN due to reduced dimensionality.

PCA on DistilBERT Embeddings

Applied to the 768-dim dense embeddings. Performance analysed across components $k \in \{10, 50, 100, 200\}$.

$$\mathbf{z}_i = W_k^T \mathbf{e}_i, \quad W_k \in \mathbb{R}^{768 \times k}$$

Finding: Sharp accuracy increase up to $k \approx 100$, then plateau — noise is filtered out while semantic structure is preserved in the top components.

3. Experiments and Models

Five classical ML models were evaluated on all feature types:

Linear Models

Logistic Regression — linear baseline, `max_iter=1000`. Optimises log-likelihood:

$$\hat{y} = \sigma(W^T \mathbf{x} + b)$$

Linear SVM — maximises the margin between classes via a linear hyperplane. Highly effective on high-dimensional sparse features.

Both models scale well with vocabulary size and are naturally interpretable via feature weights.

Non-Linear and Distance Models

RBF SVM — captures non-linear decision boundaries via the radial basis kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

KNN ($k = 5$) — distance-based; highly sensitive to the curse of dimensionality on sparse features.

KMeans ($k = 4$) — unsupervised; evaluated on its ability to recover the 4 topic classes without labels.

Evaluation Metrics:

Metrics Used

All supervised models were evaluated on: **Accuracy**, **Macro F1-score**, **Macro Precision**, and **Macro Recall**. Confusion matrices were generated for each model–feature combination to visualise class-wise predictive performance.

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot P_k \cdot R_k}{P_k + R_k} \quad \text{Accuracy} = \frac{\text{TP} + \text{TN}}{N}$$

For KMeans, cluster purity and adjusted Rand index (ARI) were used since no class labels are used during training.

4. Results and Discussion

Sparse Features Performance

Bag-of-Words Results

Logistic Regression and Linear SVM performed strongly on BoW features. Raw term frequency proved sufficient for topic classification since distinct topics use distinct vocabularies (“stock” vs. “baseball”).

KNN suffered significantly — the 5000-dim space is nearly all “equidistant” under Euclidean distance (curse of dimensionality).

Insert BoW confusion matrix / accuracy bar chart here.

TF-IDF Results

TF-IDF consistently outperformed BoW across all models. Term weighting downscales frequent uninformative words (“the”, “is”) and the bigram range captures short-range context.

Linear SVM on TF-IDF typically achieves the highest accuracy among sparse-feature models — the margin-maximisation objective pairs well with high-dimensional sparse inputs.

Insert TF-IDF confusion matrix / accuracy bar chart here.

Contextual Embeddings Performance

DistilBERT + SVM / LR

The 768-dim [CLS] embeddings provided highly separable dense representations.

SVM (RBF) and **Logistic Regression** achieved the highest overall accuracy across all feature types — capturing deep semantic meaning (synonymy, polysemy) that sparse features miss.

Insert BERT results table / confusion matrix here.

DistilBERT + KNN

KNN performed markedly *better* on BERT embeddings than on TF-IDF. Dense 768-dim vectors are continuous and compact — Euclidean distance is meaningful here, unlike in the sparse 5000-dim BoW space.

This directly illustrates the **curse of dimensionality**: distance concentration is severe in sparse high-dimensional spaces but not in dense low-dimensional ones.

Dimensionality Reduction Analysis

SVD on TF-IDF (LSA)

Reducing TF-IDF from 5000 to 100 components via Truncated SVD caused a slight drop in absolute accuracy but yielded:

- Significantly faster SVM training
- Improved KNN performance (denser space)
- Latent topic structure revealed via the singular vectors

Insert explained variance / accuracy vs. components plot here.

PCA on DistilBERT

The cumulative explained variance plot showed that the first ~100 PCA components capture the vast majority of variance in the 768-dim BERT space.

Accuracy vs. components curve showed a sharp increase from $k = 10$ to $k = 100$, then a plateau — indicating effective noise removal with minimal information loss.

Insert PCA explained variance / accuracy vs. k plot here.

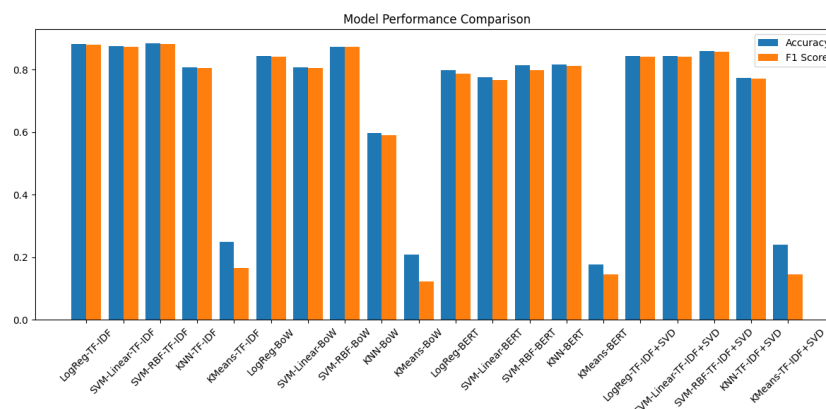


Figure 2: Accuracy vs F1 scores

5. Conclusion and Learnings

Key Takeaways

- 1. Pre-trained encoders as feature extractors:** DistilBERT [CLS] embeddings, used with zero fine-tuning, outperformed hand-crafted sparse features on all five classical models — demonstrating the power of transfer learning even in a purely statistical pipeline.
- 2. Dimensionality reduction is essential:** PCA and SVD mitigated computational bottlenecks without severely sacrificing predictive power. The accuracy plateau beyond $k = 100$ PCA components confirms that BERT embeddings are intrinsically low-dimensional.
- 3. Feature–model compatibility matters:** Linear models (LR, LinearSVM) excel on sparse high-dim features. KNN needs dense compact representations. Choosing the right feature–model pairing is as important as the model itself.